SATYAM SHARMA

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LinkedIn Portfolio

RESEARCH INTERESTS

Reinforcement learning and large language models in particular, agents that learn from past experience to make better decisions under uncertainty. Drawn to autonomous, adaptive systems and to getting capable AI working under tight compute and hardware constraints.

EDUCATION

Institute of Science Tokyo, *M.S., Industrial Engineering & Economics (Machine Learning & Reinforcement Learning)* — Apr 2026 – Present, Tokyo, Japan

Institute of Science Tokyo, *Research Student* — Oct 2025 – Mar 2026

Sant Longowal Institute of Engineering & Technology (SLIET), *B.E. (Honours), Computer Science & Engineering.* — Jul 2021 – Jun 2025, 8.9 / 10 CGPA

RESEARCH

Graduate Researcher, *Ichise Laboratory, Institute of Science Tokyo* — Oct 2025 – Present

- Exploring reinforcement learning for agents that learn from past experience to improve decision-making under uncertainty, working at the intersection of RL and large language models.
- Survey and present recent RL and LLM literature in regular lab seminars while shaping a thesis direction.

PROJECTS

Earth Docker, *Autonomous Agricultural Robot*

- Won competitive project funding as the only second-year student selected, alongside final-year seniors.
- Trained a ResNet152V2 classifier to 98% accuracy distinguishing diseased from healthy cotton plants and leaves.
- Deployed the full perception-and-navigation stack on a Raspberry Pi 4 with RP-LiDAR for path planning capable AI on low-cost, low-power hardware.
- Designed a rocker-bogie chassis with planetary-gear drive to handle uneven farm terrain.

EduMind AI , *LLM-Powered Study Platform*

- Built a three-tier web app (React/Material-UI; Node/Express; MongoDB) integrating Ollama-served LLaMA 3.1/3.2 and Tesseract OCR for real-time Q&A, adaptive mind-maps, study plans, and automated essay scoring (typed and handwritten) across 8+ competitive exams.
- Engineered asynchronous queues, prompt caching, and proxy optimization response.

NLP Portfolio
- Built a multi-class emotion-detection model (TF-IDF, n-grams, Random Forest / Naive Bayes) reaching 94% accuracy with careful text preprocessing.
- Scraped and processed 10,000+ company records ratings, reviews, salaries, listings from AmbitionBox with requests and BeautifulSoup.

SCHOLARSHIPS & ACHIEVEMENTS

MEXT (Monbukagakusho) Scholarship,
Ministry of Education, Culture, Sports, Science and Technology (MEXT) of Japan
- One of 25 Indian nationals awarded in 2025 fully funded for graduate study at the Institute of Science Tokyo.
- Scholarship Recipient for Master's studies at the Institute of Science, Tokyo.

B.E. in Computer Science and Engineering with Honours distinction, *(8.9 CGPA).*
Completed B.E in Computer Science and Engineering along with Honours degree as well.

LEADERSHIP

Robotics Club, SLIET , *Convenor* Feb 2022 – Jul 2025
Led projects spanning drones, line-following robots, and hovercraft represented the club at IIT-BHU's Technex-23 technical fest.

TechFest23 (Robozar domain), *Event Coordinator*
Planned and ran inter-college robotics events end to end, managing participation from teams across colleges.

TECHNICAL SKILLS

Languages: Python, C/C++, JavaScript, HTML/CSS
ML / AI: TensorFlow, Keras, PyTorch, Scikit-Learn, NumPy, Pandas, Matplotlib; MLflow, Prefect
Focus areas: Reinforcement Learning, Deep Learning, NLP, Computer Vision
Tools & systems: Git/GitHub, Linux, Raspberry Pi & embedded deployment, REST APIs

EXTRA-CURRICULAR

- Participant, IRSO Robotic Challenge (URSC), Robotics design and engineering challenge.
- Active participant, IIT-BHU Techfest 2023.

CERTIFICATIONS

Machine Learning Specialization , *Andrew Ng, Coursera*

Natural Language Processing , *Coursera*